

Supplementary Material

1. Description of the collected variables

Table 1: List of the adopted variables to represent the flood economic losses and claims

Variable	Code	Variable Description
Quartile losses	FE1	Defined quartile losses based on the event intensity level
Province	FE2	Name of the affected province
Month	FE3	The month when the disaster occurred
Year	FE4	The year when the disaster occurred
Number of claims	FE5	Number of claims related to the previous year of the event
Total cost	FE6	Total cost for the same period but in the previous year
Duration	FE7	Event duration related to the previous year of the event
Municipalities	FE8	Number of affected municipalities based on the previous year
Neighbourhoods	FE9	Number of affected neighbourhoods based on the previous year
Affected houses	FE10	Number of affected houses based on the previous year
Affected transportation	FE11	Number of affected transportation (e.g. cars, trucks, trains, etc) on the previous year
Affected stores	FE12	Number of affected stores on the previous year
Affected industries	FE13	Number of affected industries on the previous year
Affected offices	FE14	Number of affected offices on the previous year
Affected constructions	FE15	Number of affected constructions on the previous year

Table 2: List of the adopted variables to represent the meteorological characteristics

Variable	Code	Format	Variable Description
Max temperature	M1	Celsius (C)	The maximum temperature during the event.
Min temperature	M2	Celsius (C)	The minimum temperature during the event.
Avg. temperature	M3	Celsius (C)	The average temperature during the event.
Avg. wind speed	M4	Km/h	The wind velocity during the event.
Max wind speed	M5	Km/h	The maximum wind speed during the event.
Precipitation	M6	Millimetres (mm)	The registered precipitation during the event.

Table 3: List of adopted variables to represent the resilience level of the region. This representation illustrates the variables associated with the BRIC framework, their descriptions, and the relevant literature background.

BRIC Category	Index	Variable	Variable Description	Justification/ Inspiration
Social	RS1	Total population	Quantity of citizens in the province	(Heinzle et al., 2020; Pazhuhan et al., 2023)
	RS2	Median age	Measure the median age of the population	(Heinzle et al., 2020; Singh-Peterson et al., 2014)

	RS3	Population until 15	Quantity of the young population	(Heinzlef et al., 2020; Pazhuhan et al., 2023; Singh-Peterson et al., 2014)
	RS4	Population over 65	Quantity of the population that reflects elderly individuals	(Heinzlef et al., 2020; Pazhuhan et al., 2023; Singh-Peterson et al., 2014)
Economic	REc1	Average net income per person	Represents the average individual income in Euros	(Javadpoor et al., 2021)
	REc2	PIB per capita	Measures the economic output per person in Euros	(Javadpoor et al., 2021)
	REc3	Population that is of working age (16 – 64 years)	Quantity of the economic segment of the population	(Javadpoor et al., 2021)
	REc4	Income household	Indicates the average household income in Euros	(Javadpoor et al., 2021; Singh-Peterson et al., 2014; Stefano et al., 2017)
	REc5	Average salary	Reflects the average earnings in euros	(Javadpoor et al., 2021; Singh-Peterson et al., 2014)
	REc6	Income less than 10,000	Represents the percentage of the population earning low incomes	(Javadpoor et al., 2021; Singh-Peterson et al., 2014)
	REc7	Hours worked per person	Measures the average work hours	(Javadpoor et al., 2021)
	REc8	Employment	Indicates the percentage of the population that is employed	(Opach et al., 2020; Scherzer et al., 2019; Singh-Peterson et al., 2014)
	REc9	Spending household	Reflects household consumption patterns in Euros	(Javadpoor et al., 2021; Singh-Peterson et al., 2014)
	REc10	Industry	Indicates the percentage of the workforce in industrial jobs	(Scherzer et al., 2019)
	REc11	Construction	Indicates the percentage of the workforce in construction jobs	(Scherzer et al., 2019)
	REc12	Service	Indicates the percentage of the workforce in service jobs	(Scherzer et al., 2019)
	REc13	Number of businesses	Represents the total number of businesses	(Scherzer et al., 2019)
	REc14	Industrial businesses	Indicates the number of businesses in the industrial sector	(Scherzer et al., 2019)

	REc15	Construction businesses	Represents the number of businesses in the construction sector	(Scherzer et al., 2019)
	REc16	Commercial, transport, and hospitality businesses	Reflects the number of businesses in the commercial, transport, and hospitality sectors	(Scherzer et al., 2019)
Community Capital	RC1	Net school enrolment rate between 18 and 22	Percentage of young adults in the education	(Stefano et al., 2017)
	RC2	Mandatory studies between 25 and 64	Percentage of the education level of adults	(Stefano et al., 2017)
	RC3	Spanish population	Quantity of the native population	(Pazhuhan et al., 2023)
	RC4	Foreign population	Quantity of immigrant population	(Pazhuhan et al., 2023)
Institutional	RI1	Unemployment	Reflects the percentage of the workforce without jobs	(Scherzer et al., 2019)
	RI2	Thousands of households in property	Represents homeownership rates	(Singh-Peterson et al., 2014; Stefano et al., 2017)
Housing/ Infrastructure	RH1	Road extension	Represents the length of road networks in KM	(Opach et al., 2020; Scherzer et al., 2019)
	RH2	Coast extension	Reflects the length of coastal areas KM	(Opach et al., 2020; Scherzer et al., 2019)
	RH3	Thousands of households	Reflects the number of housing units	(Heinzlef et al., 2020; Pazhuhan et al., 2023; Singh-Peterson et al., 2014)
	RH4	People per households	Indicates the average household size	(Heinzlef et al., 2020; Pazhuhan et al., 2023; Singh-Peterson et al., 2014)
Environmental	REn1	Agriculture surface	Represents the area used for agriculture in KM	(Opach et al., 2020; Scherzer et al., 2019)
	REn2	Total surface	Reflects the total land area of the region in KM	(Scherzer et al., 2019)
	REn3	Agriculture	Indicates the percentage of the workforce in agriculture	(Scherzer et al., 2019)

Table 4: List of governmental interventions promoted by the Spanish government at different administration levels. The data were extracted from monitoring plans from the flood management action.

Phase	Main Measure	Code	Indicator
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Prevention	Land-use planning and urban development (LUPUD)	13.01.01.01	Number of urban planning reports issued by the Basin Organisations in relation to Article 25.4
		13.01.01.02	Average time for the issuance of urban planning reports by the Basin Organisations in relation to Article 25.4
		13.01.01.03	Number of protocols, agreements and other agreements signed with competent administrations
		13.01.01.04	Percentage of regulations in the Autonomous Communities coordinated with floods
		13.01.01.05	No. of urban planning under review
		13.01.01.06	No. of updated urban planning
		13.01.01.07	Number of cartographic viewers on the internet with information on flooding
		13.01.01.08	No. of guides and technical manuals developed on construction criteria to reduce the vulnerability of exposed elements in flood zones
		13.01.01.09	No. of removal/relocation actions in flood zones
		13.01.01.10	Budget for removal/relocation in flood zones
		13.01.01.11	Number of urban planning reports issued by the AGE in relation to articles 222 and 227 of the General Coastal Regulation
	Measures to improve knowledge on flood risk management (FRKM)	13.04.01.01	Km of riverbed with flood zone mapping
		13.04.01.02	Number of institutions present in the flood R+D+i interest group
		13.04.01.03	Number of projects submitted to R+D+i calls on floods
		13.04.01.04	Number of projects selected in the R+D+i calls on floods
		13.04.01.05	Status of studies defining the magnitude and frequency of floods (and coastal storms)
		13.04.01.06	Status of studies on the effects of climate change on floods
		13.04.01.07	Km of coastline with flood zone mapping
	Programme for the maintenance and conservation of watercourses (WCON)	13.04.02.01	No. of actions for the maintenance and conservation of riverbeds by responsible body
		13.04.02.02	Km of riverbed subject to action
		13.04.02.03	Annual investment in maintenance and conservation of riverbeds
		13.04.02.04	Status of the drafting and implementation of the Good Practices Manual.
	Monitoring of the programme for the maintenance and conservation of the coastline (CTSM)	13.04.03.01	Number of coastal maintenance and conservation actions by responsible body
		13.04.03.02	Km of coastline subject to action

Protection		13.04.03.03	Annual investment in coastal maintenance and conservation
		13.04.03.04	No. of coastal replanting and reforestation actions
		13.04.03.05	M3 of sediment contributed to or removed from beaches or dunes
	Measures in the basin: Hydrological-forestry restoration and agro-hydrological management (HFRM)	14.01.01.01	No. of forest agrohydrological restoration projects by responsible body
		14.01.01.02	Annual investment in forest agrohydrological restoration
		14.01.01.03	Soil area (ha) on which forest agrohydrological restoration is carried out
	Measures in the river channel and floodplain: River restoration, including natural water retention measures and riparian reforestation (RSTR)	14.01.02.01	Number of river restoration actions by responsible body
		14.01.02.02	Km of setback of longitudinal fenders
		14.01.02.03	Km of recovery of the layout of old riverbeds
		14.01.02.04	Km of recovered riverbed
		14.01.02.05	Length of water bodies river typology where fluvial restoration is carried out
		14.01.02.06	No. of river restoration actions that include measures for natural water retention executed/in execution
		14.01.02.07	Km of riverbed subject to river restoration action in ARPSIs
		14.01.02.08	Number of agreements/agreements signed for river restoration in ARPSIs
		14.01.02.09	Annual investment in river restoration in the ARPSIs
		14.01.02.10	No. of transverse barriers removed
		14.01.02.11	Number of adapted barriers for fish migration
		14.01.02.12	Km of river connected by adaptation/removal of transverse barriers
		14.01.02.13	Km of removal of longitudinal fenders
	Measures to improve drainage of linear infrastructures: Roads, railways (LINF)	14.01.03.01	No. of coastal restoration projects
		14.01.03.02	No. of coastal replanting and reforestation actions
		14.01.03.03	No. of conventions/agreements signed for coastal restoration in ARPSIs
		14.01.03.04	Annual investment in coastal restoration in the ARPSIs
		14.01.03.05	Km of coastline subject to coastal restoration action in ARPSIs
	Rules for managing the operation of reservoirs that have a significant impact on the hydrological regime	14.02.01.01	Percentage of large state-owned dams with approved operating rules

	(RESR)	14.02.01.02	Percentage of large concessionaire dams with approved operating standards
	Structural measures to regulate flows, such as the construction and/or modification of dams exclusively for flood defence (DAMS)	14.02.02.01	Status of the drafting of the technical guide for carrying out cost-benefit studies
		14.02.02.02	No. of new dams (under construction or executed) included in the Flood Risk Management Plan intended only for flood lamination
	Measures to improve drainage of linear infrastructures: Roads, railways (DRLI)	14.03.01.01	No. of transverse drainage improvement works executed/in execution
		14.03.01.02	Total investment in works to improve the transversal drainage by each competent administration
		14.03.01.03	Status of the update of the transverse drainage regulations
	Structural measures (channelisation, levees, dikes, etc.) that involve physical interventions in river channels, coastal waters, and flood-prone areas (STRC)	14.03.02.01	Status of the drafting of the technical guide for carrying out cost-benefit studies
		14.03.02.02	km of new pipelines (under execution or executed) included in the Flood Risk Management Plan
		14.03.02.03	km of new mounds or (under construction or executed) included in the Flood Risk Management Plan
		14.03.02.04	Investment for the different works planned
		14.03.02.05	Km of new coastal works (coastal walls, dikes, breakwaters, freestanding dikes, movable barriers)
		14.03.02.06	m2 of land reclaimed from the sea by filling
	Measures involving physical interventions to reduce flooding by surface waters, generally, although not exclusively, in an urban environment, such as improving artificial drainage capacity or sustainable drainage systems (SUDS)	14.04.01.01	No. of regulatory provisions or guides to good practice promoted by the competent Administrations in the municipalities that include ARPSISs
		14.04.01.02	Evolution of the number of urban centres with SuDS implemented in relation to the ARPSIs and flood risk
Preparation	Measures to improve meteorological warning systems (METW)	15.01.01.01	Meteoalerta update and upgrade status
		15.01.01.02	Number of Meteoalerta activations related to the flood protocol
		15.01.01.03	Status of coordination with hydrological warning systems
	Measures to establish or improve hydrological measurement and warning systems (HYDW)	15.01.02.01	Number of measurement and control points available
		15.01.02.02	Annual investment for network integration, operation and maintenance
		15.01.02.03	No. of activations of the Hydrological Alert Protocol
	Measures to establish or improve institutional planning of	15.02.01.01	No. of approved regional civil protection plans

	emergency response to floods through coordination with Civil Protection Plans (EMPL)	15.02.01.02	Number of existing Civil Protection Plans updated in accordance with the content of the PGRI
		15.02.01.03	Number of local action plans drawn up
	Measures to improve action protocols and communication of information related to floods (PROT)	15.02.02.01	No. of bodies involved in the Communication Protocol
	Measures to establish or improve public awareness in preparation for floods, increasing the perception of flood risk and self-protection strategies among the population (AWRN)	15.03.01.01	Number of conferences and training campaigns held among the various social actors and administrations involved in raising public awareness of flood risk
		15.03.01.02	No. of administrations that incorporate information on flood risks on their websites
		15.03.01.03	Number of meetings and writings for dissemination of PLANINPA among action groups
		15.03.01.04	Number of municipal action protocols carried out and implemented
		15.03.01.05	Result of the risk perception survey
		15.03.01.06	Number of conferences and training campaigns held with the population and social agents of risk areas
Recovery/Review	Emergency works for the repair of affected infrastructure, including basic health and environmental infrastructure (REPR)	16.01.01.01	Number of episodes classified as "serious floods"
		16.01.01.02	Number of emergency actions carried out
		16.01.01.03	Annual budget invested in emergency actions
		16.01.01.04	Number of requests for aid (General Directorate of Civil Protection and Emergencies)
		16.01.01.05	Budget of the aid granted (General Directorate of Civil Protection and Emergencies)
	Civil Protection Plans: Support actions for health, financial assistance, including legal aid, as well as temporary relocation of the affected population (CPSP)	16.01.02.01	No. of actions for the temporary relocation of the population carried out following the Civil Protection plans linked to floods
		16.01.02.02	Number of people affected in flood episodes during the period
		16.01.02.03	Damage caused by flood episodes that occurred during the period
	Promotion of flood insurance for people and property, including agricultural insurance (INSU)	16.03.01.01	Evolution of the insured capital in extraordinary risks
		16.03.01.02	Evolution of the degree of satisfaction of policyholders.
		16.03.01.03	Evolution of the annual loss rate paid in floods
		16.03.01.04	Evolution of the number of agricultural insurance policies
		16.03.01.05	Annual amount of subsidies applied by ENESA for the subscription of

			agricultural insurance to farmers and livestock breeders
		16.03.01.06	Annual amount of compensation paid in floods to insured persons under the agricultural insurance system
	Measures for evaluation, analysis, and diagnosis of lessons learned from flood event management (EVAL)	16.03.02.01	No. of evaluation reports prepared
		16.03.02.02	Number of technical conferences for the dissemination of lessons learned

2. Steps to clean and select the variables for the economic loss prediction

